FORMATIVE ASSESSMENT -1
PROJECT WORK

1. NAME OF THE PROJECT

Investigating the Properties and Real-Life Usages of Concave and Convex Mirrors

2. PURPOSE OF THE PROJECT

The primary purpose of this project is to practically analyze the formation of images by spherical mirrors (concave and convex) when an object is placed at varying distances. Furthermore, this project aims to correlate these optical properties—such as magnification and field of view—with their respective real-world applications and usages.

3. HYPOTHESIS

It is hypothesized that a **concave mirror** will produce both real (inverted) and virtual (erect) images depending on the distance of the object from the pole, allowing it to act as a magnifier or a light concentrator. Conversely, a **convex mirror** will always produce virtual, erect, and diminished images regardless of object position, thereby providing a significantly wider field of view.

4. MATERIALS USED

- A Concave mirror of known focal length (e.g., $f = 15\ cm$)

- A Convex mirror
- V-stand (Mirror holder)
- Meter scale (for measuring distances)
- A lighted candle (acting as the object)
- White stiff cardboard (acting as the screen)
- A semi-dark room to clearly observe image formation

5. PROCEDURE

1. Mount the concave mirror on the V-stand and place it on a long table.
2. Determine the approximate focal length of the mirror by focusing a distant object (like a tree outside the window) onto the white screen until a sharp point is obtained. Measure this distance.
3. Place the meter scale along the principal axis of the mirror. Mark the focal point (F) and the center of curvature (C) on the table.
4. Place the lighted candle at different positions: Beyond C, At C, Between C and F, At F, and Between F and the Pole (P).
5. For each position (except when the image is virtual), move the screen back and forth until a sharp, distinct image of the candle flame is captured.
6. Note down the position, size (magnified, diminished, or same size), and nature (real/virtual, erect/inverted) of the image.
7. Replace the concave mirror with a convex mirror and attempt to catch the image on the screen. Observe the image directly in the mirror to record its nature.

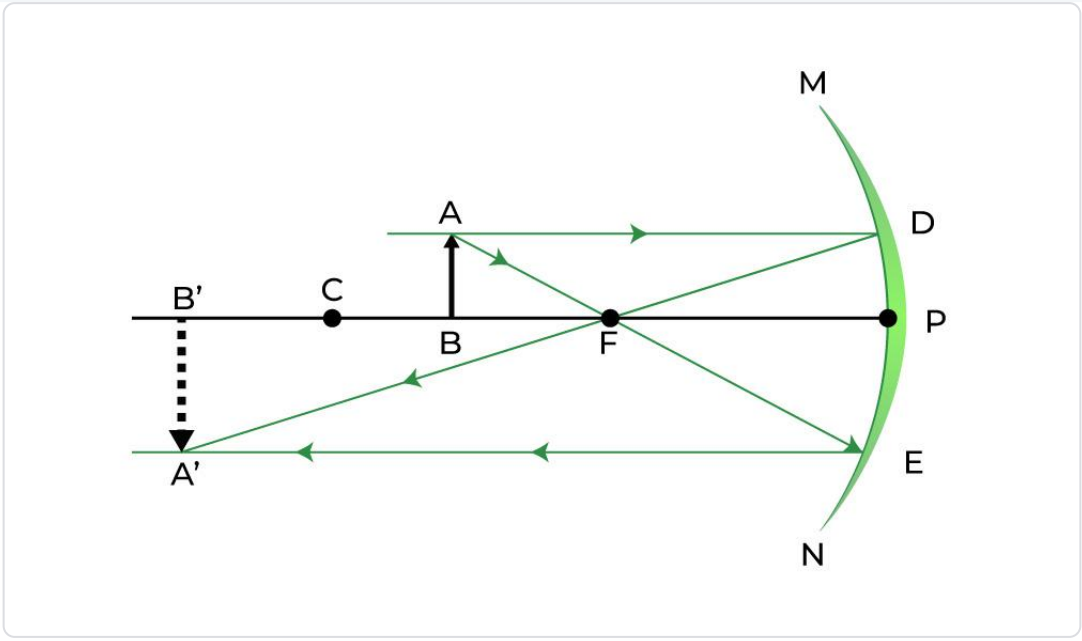


Image 1: Accurate Paraxial Ray Diagram showing a Real, Inverted, and Magnified image formed by a Concave Mirror (Object between C and F)

6. DATA COLLECTION

The following table records the experimental observations for the Concave Mirror at various object positions:

Position of the Object	Position of the Image	Size of the Image	Nature of the Image
At Infinity	At Focus (F)	Highly Diminished (Point sized)	Real and Inverted
Beyond Center of Curvature (C)	Between F and C	Diminished	Real and Inverted
At Center of Curvature (C)	At C	Same Size	Real and Inverted
Between C and F	Beyond C	Magnified	Real and Inverted
At Focus (F)	At Infinity	Highly Magnified	Real and Inverted
Between F and Pole (P)	Behind the Mirror	Magnified	Virtual and Erect

For Convex Mirror: Irrespective of the object's position, the image was always formed *behind the mirror*, and was consistently **Virtual, Erect, and Diminished**.

7. INTRODUCTION

Spherical mirrors are reflecting surfaces that are a part of a hollow sphere of glass. The reflection of light at these curved surfaces follows specific geometric rules. There are two primary types of spherical mirrors:

- **Concave Mirror:** The reflecting surface curves inwards, towards the center of the sphere. It is generally a *converging mirror*.
- **Convex Mirror:** The reflecting surface curves outwards. It is generally a *diverging mirror*.

Understanding the focal length (***f***), pole (P), center of curvature (C), and how light rays interact with these curved surfaces is essential in modern optics and daily life applications.

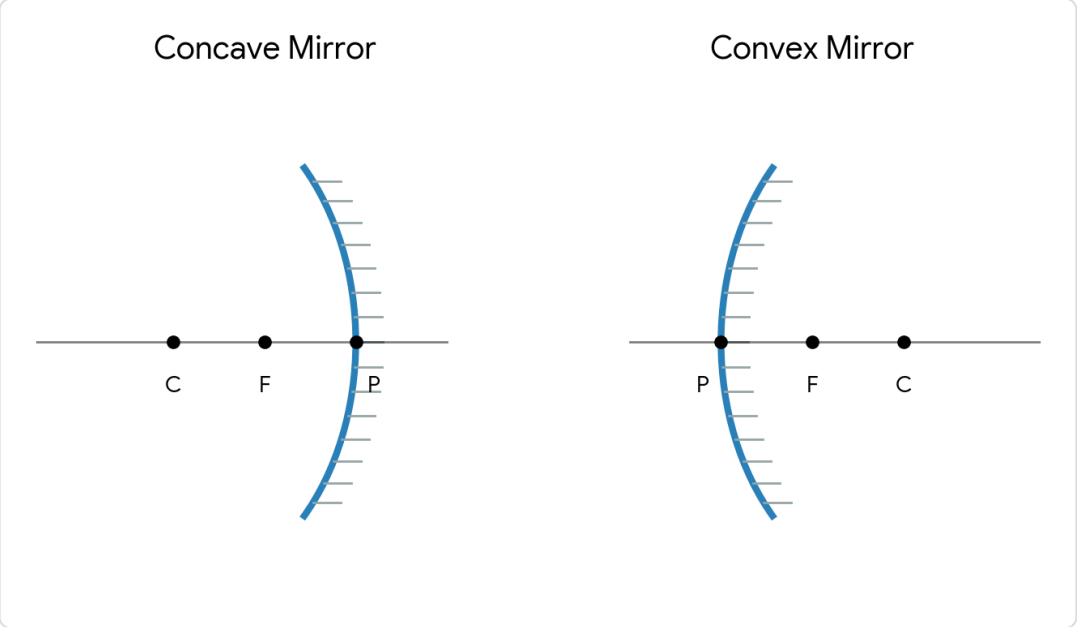


Image 2: Geometric profiles of Concave and Convex Spherical Mirrors

8. USES OF CONCAVE MIRROR

Based on their converging and magnifying properties, concave mirrors are utilized extensively in everyday life:

- **Vehicle Headlights and Torches:** When a light bulb is placed at the principal focus of a concave mirror, the reflected light travels as a powerful, parallel beam spanning long distances.
- **Dentist's Mirrors:** Dentists use small concave mirrors to see a magnified, erect, virtual image of the patient's teeth to detect cavities easily.
- **Shaving and Makeup Mirrors:** When the face is held very close to the mirror (between the pole and focus), a magnified and upright image is formed, aiding in grooming.
- **Solar Furnaces:** Large concave mirrors are used to converge sunlight at a single focal point, producing immense heat suitable for industrial melting processes.

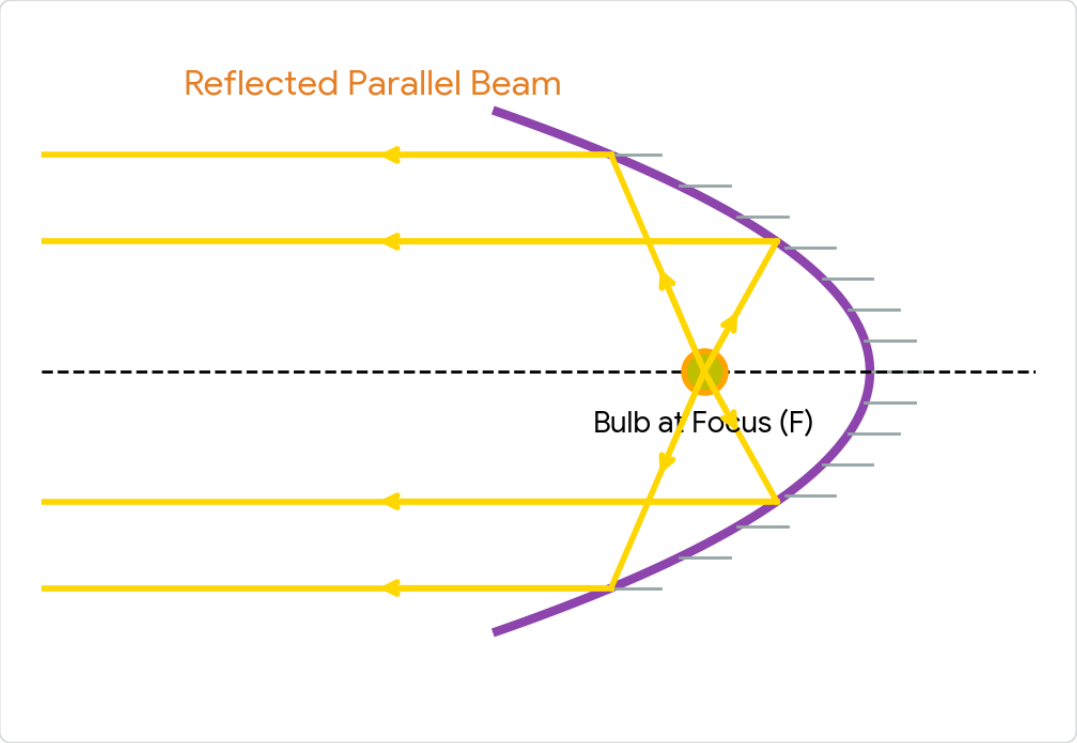


Image 3: Application of a Concave Mirror as a Headlight Reflector (Parallel Beam Production)

9. USES OF CONVEX MIRROR

Because convex mirrors always form erect and diminished images, they are excellent at expanding the visual field. Their primary uses include:

- **Rear-View Mirrors in Vehicles:** Drivers use convex mirrors mounted on the sides of cars and motorcycles. They provide a much wider field of view compared to plane mirrors, helping the driver see traffic approaching from behind safely.
- **Safety and Surveillance Mirrors:** Large convex mirrors are installed at blind corners of narrow roads, in parking garages, and in supermarkets to prevent collisions and monitor large areas.
- **Street Light Reflectors:** They are used to diverge light over a larger area on the streets.

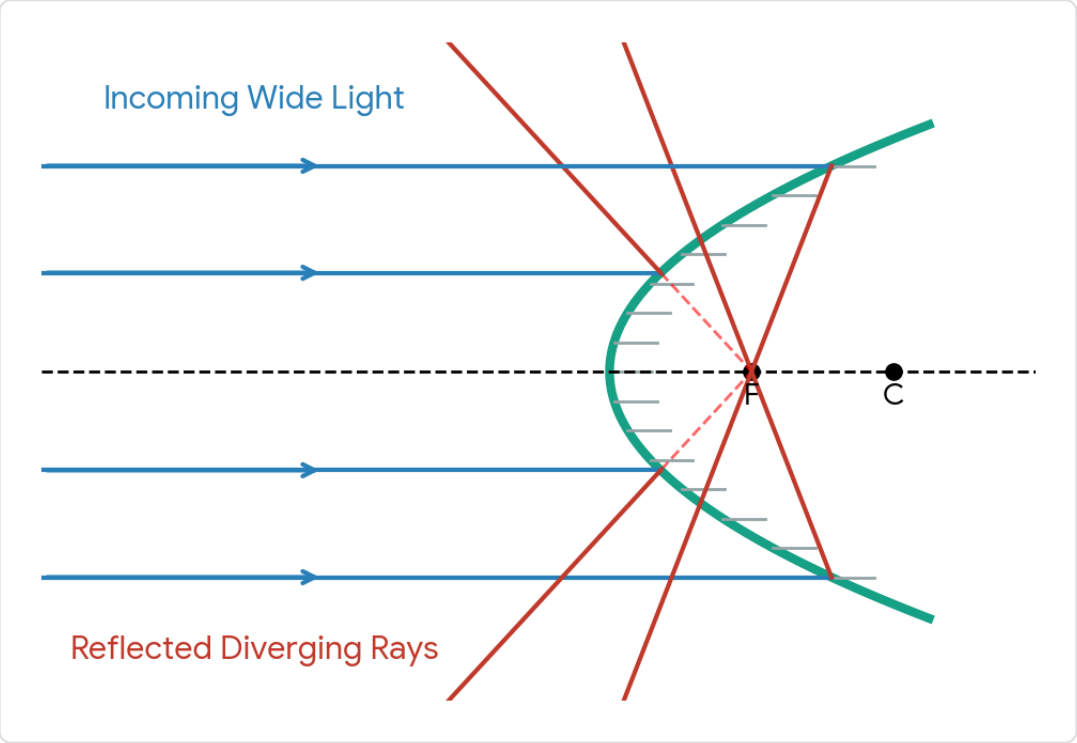


Image 4: Application of a Convex Mirror as a Rear-View Mirror (Wide Field of View)

10. OBSERVATION

From the collected data, it is clearly observed that the concave mirror creates real and inverted images that can be captured on a screen for most object positions. As the object moves closer to the mirror (from infinity to the focus), the image moves further away and increases in size. Once the object crosses the focal point (F), the rays diverge, and a virtual, magnified, and erect image is seen inside the mirror.

In contrast, the convex mirror never forms a real image. The reflected rays continuously diverge, creating a small, upright virtual image that allows the observer to see a vast area behind them compressed into a small mirror surface.

11. CONCLUSION

The project successfully demonstrates the fundamental principles of light reflection at curved surfaces. The initial hypothesis was validated through practical experimentation and data

collection. Concave mirrors act as versatile optical tools that can converge light beams and magnify images, which makes them ideal for headlights, solar energy concentration, and medical instruments. Convex mirrors, serving as diverging surfaces, are crucial for vehicular safety and surveillance by providing an extended field of view. Understanding these properties enables us to design and utilize appropriate mirrors for specific technological and daily-life requirements.

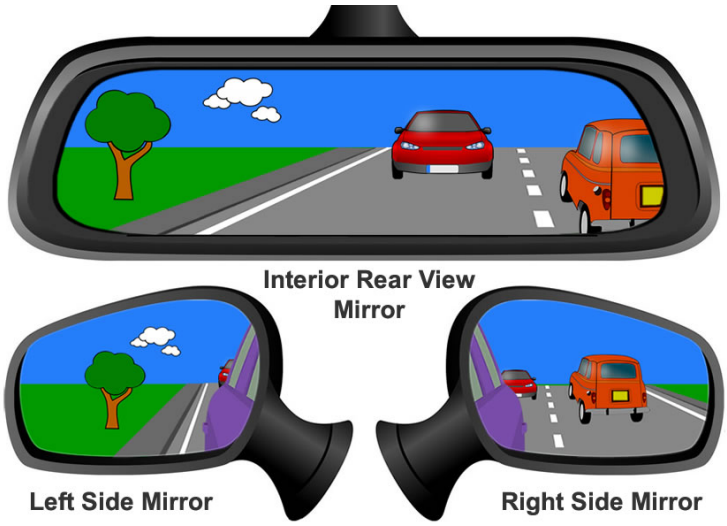











Interior Rear View
Mirror
Left Side Mirror
Right Side Mirror

